Unit 2 Classwork — Exponent Rules — Answer Key

Exponent Rules · Power/Zero/Negative Exponents

A. Exponent Rules — Product & Quotient

Problem 1 Score 3

Simplify: $x^4 \cdot x^3$

Answer: x^7

Problem 2 Score 3

Simplify: $y^8 \div y^3$

Answer: y^5

Problem 3 Score 4

Simplify: $(2x^3)(3x^2)$

Answer: $6x^5$

Problem 4 Score 4

Simplify: $15a^6 \div 3a^2$

Answer: $5a^4$

Problem 5 Score 4

Simplify: $a^2 \cdot a^5 \cdot a$

Answer: a^8

B. Power, Zero & Negative Exponents

Problem 1 Score 3

Simplify: $(x^3)^4$

Answer: x^{12}

Problem 2 Score 3

Simplify: 7^0

Answer: 1

Problem 3 Score 4

Simplify: $(2x^2)^3$

Answer: $8x^6$

Problem 4 Score 4

Simplify: $(5x^4)^0$

Answer: 1

Problem 5 Score 4

Simplify: 3^{-2} (write as a fraction)

Answer: $\frac{1}{9}$

Problem 6 Score 4

Simplify: x^{-4} (write as a fraction)

Answer: $\frac{1}{x^4}$

Problem 7 Score 5

Evaluate: -4^2 vs $(-4)^2$ — what is $(-4)^2$?

Answer: 16

Problem 11 Score 5

Which shows why $5^0 = 1$?

A. $5 \times 0 = 0$, and $0 + 1 = 1$.

B. $5^1 = 5$, so 5^0 is $5 - 4 = 1$.

C. It is only a rule to memorize. There is no reason.

D. $5^3 \div 5^3 = 5^0$, and a number divided by itself is 1.

Answer: D

Problem 12 Score 5

Simplify: $(2x^3y)^2 \cdot (3x^{-4}y^5)$

A. $12x^2y^7$

B. $6x^2y^7$

C. $12x^{10}y^7$

D. $12x^2y^{10}$

Answer: A

Problem 13 Score 5

Simplify: $(4a^5b^2)(3a^{-2}b) \div (6a^3b^4)$

A. $6/b$

B. $2a^6/b$

C. $2/b$

D. $2b$

Answer: C